Pinned group summary: SU(n=2, q=1, eps=-1)

Pinning-Sympy automated output

July 8, 2026

Contents

1 Basic group attributes

Group	SU(n=2, q=1, eps=-1)
Matrix size	2×2
Rank	1
Defining equations	$DefiningEquationPlaceholder$

1.1 Skew-hermitian form

Dimension	2
Witt index	1
Primitive element	$\sqrt{d}$
Epsilon	-1

Matrix: $\begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$

1.2 Lie algebra

The defining equations for the Lie algebra are:

$$LieAlgebraEquationPlaceholder$$

A generic element of the Lie algebra is:

$$\begin{bmatrix} x_{r00} & -x_{r01} \\ -x_{r10} & -x_{r00} \end{bmatrix}$$

1.3 Torus

The torus has rank 1. A generic element of the torus is

$$\begin{bmatrix} t_0 & 0 \\ 0 & \frac{1}{t_0} \end{bmatrix}$$

The rows of the matrix below are trivial characters:

$$\begin{bmatrix} 1 & 1 \end{bmatrix}$$

2 Root system

Dynkin type	A_1
Irreducible	True
Reduced	True
Simply laced	True
Number of roots	2

2.1 Dynkin diagram

The root system is irreducible and has Dynkin diagram:

2.2 Cartan matrix

The root system is irreducible and has Cartan matrix:

$$\begin{bmatrix} 2 \end{bmatrix}$$

2.3 Roots

Root	**Simple**	**Sign**	**Multipliable**
$(-2,\ 0)$	No	$-$	No
$(2,\ 0)$	Yes	$+$	No

2.4 Coroots

Root	**Coroot**
$(-2,\ 0)$	$(-1,\ 1)$
$(2,\ 0)$	$(1,\ -1)$

2.5 Linear combinations of roots

Definition 2.1. For two roots α, β in a root system Φ, the set of positive integral linear combinations of α, β that are also roots is denoted $(\alpha, \beta) = \{i\alpha + j\beta \in \Phi : i, j \in \mathbb{Z}_{\geq 1}\}$.

No pairs of roots generate positive integer linear combinations.

3 Root spaces

3.1 Dimensions

The table below lists each root along with the dimension of the associated root space (and root subgroup).
RootSpaceDimensionPlaceholder

3.2 Root space table

The table below lists each root along with a generic element of the associated root space (inside the Lie algebra). Include a definition here? INCOMPLETE
RootSpaceTablePlaceholder

4 Root subgroups

4.1 Root subgroup table

The table below lists each root along with a generic element of the associated root subgroup.
RootSubgroupTablePlaceholder

4.2 Homomorphism property

Background theoretical info on homomorphism/pseudo-homomorphism property INCOMPLETE
HomomorphismDefectTablePlaceholder
Below is an explicit formulation of the homomorphism/pseudo-homomorphism identities.
HomomorphismDefectIdentitiesPlaceholder

5 Commutators

Background theoretical info on commutator relation INCOMPLETE
Because of its relevance to commutators, we repeat here the table of linear combinations of roots.
No pairs of roots generate positive integer linear combinations.

5.1 Commutator coefficient table

The table below gives the coefficients $N_{ij}^{\alpha\beta}$.
CommutatorCoefficientTablePlaceholder

5.2 Commutator identities

Below is an explicit formulation of the commutator formula identities.
CommutatorIdentitiesPlaceholder

6 Weyl group

Background theoretical info on Weyl group, especially as realized inside the group, including info on conjugation between root subgroups INCOMPLETE

6.1 Weyl group elements

The table below lists roots along with an associated Weyl group element.

WeylGroupPlaceholder

6.2 Weyl group conjugation coefficients

The table below lists roots along with the coefficients obtain in performing conjugation by an associated Weyl element.

WeylConjugationCoefficientPlaceholder

6.3 Weyl group conjugation identities

Below is an explicit formulation of the Weyl group conjugation identities.

WeylConjugationIdentitiesPlaceholder